

INFORMATION SYSTEMS & BUSINESS INTELLIGENCE

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1. Introduction

1.1 BACKGROUND INFORMATION

In today's data-driven world, understanding and leveraging analytics for environmental and ecological monitoring is becoming increasingly critical. Our project aimed to harness the power of data analytics to gain insights into environmental patterns and insect activity using two interconnected datasets:

1. **Metro Data:** Includes temperature, low temperature, high temperature, humidity, and timestamps.
2. **Captured Insects Data:** Contains information about the total number of insects, new captures, and reviewed insects over allocated dates and times.

To achieve this, we implemented an **Exploratory Data Analysis (EDA)** using **Streamlit**, a versatile and user-friendly framework for data science and machine learning applications. The EDA process involved computing missing values, visualizing data distributions, and identifying relationships between variables in both datasets.

Following the EDA, we applied various modelling techniques to derive meaningful insights and actionable conclusions. These techniques included:

1. **Linear Regression:**
Utilized to model the relationship between temperature, humidity, and the number of insects captured, allowing us to predict insect activity based on weather conditions.
2. **ARIMA:**
Deployed for time series analysis to forecast trends and patterns in insect captures and environmental factors over time, enabling proactive planning and monitoring.
3. **K-Means Clustering:**
Used to group similar patterns of insect activity and weather conditions, helping identify clusters of high insect activity periods and corresponding environmental conditions.

This comprehensive approach aimed to provide a detailed analysis to inform data-driven decisions in environmental and ecological research. By integrating insights from weather and insect activity data, we hope to enhance understanding and support strategies to address challenges such as pest control, biodiversity monitoring, and ecological balance.

1.2 SIGNIFICANCE AND MOTIVATION

The significance and motivation of using the approach outlined in our project lie in several key factors:

□ **Comprehensive Data Analysis:**

By utilizing historical datasets, including environmental (Metro Data) and ecological (Captured Insects Data) information, and processing them with advanced data analysis techniques, we gain insights into complex relationships between temperature, humidity, and insect activity. This enables the identification of trends, patterns, and anomalies that may not be apparent through manual analysis.

□ **User-Friendly Data Exploration:**

Implementing **Streamlit** for visualization and analysis creates an interactive and accessible platform for data exploration. This approach empowers users to interact with the data intuitively, facilitating deeper insights without requiring extensive technical expertise.

□ **Advanced Analytics:**

By employing modelling techniques like Linear Regression, ARIMA, and K-Means Clustering, we uncover significant relationships and trends within the data. These techniques enable predictions of insect activity based on environmental conditions, forecasting future trends, and identifying clusters of similar conditions for targeted insights.

□ **Efficient Data Handling:**

Using streamlined data preprocessing and visualization techniques ensures efficient handling of datasets, enabling in-depth analysis even with substantial data volumes. This approach eliminates hardware constraints and focuses on practical, scalable solutions for environmental and ecological analysis.

□ **Holistic Approach:**

By integrating exploratory data analysis (EDA) and predictive modelling into a unified framework, our project provides a comprehensive solution for understanding the relationships between environmental conditions and insect activity. This approach delivers actionable insights that can support decision-making in areas such as pest control, environmental monitoring, and ecological research.

1.3 METHODOLOGIES

To deal with the aims and objectives of the project we chose to use the following tools and technologies for our project. They are as follows:

1.3.1 HARDWARE

The development process was done by the following laptops as shown:

Description	Configuration
Laptops	HP Omen 16

1.3.2 SOFTWARE

The development process was done by the following operating systems and software as shown:

Description	Technology
Operating Systems	Windows 11
Editor	Google Collab, Jupyter Notebook

1.3.3 DEVELOPMENT SIDE TECHNOLOGIES

The development process was done by the following technologies as shown:

Description	Technology
Framework	Streamlit
Collaboration	Local host
Summarizer Engine	VS Code v1.86.1
UML Diagrams	Star UML v5.0.1

2. Literature Review

2.1 INTRODUCTION

Since the inception of this project, we have taken everything under consideration with respect to research. From gathering the prerequisites of the project to deploying it, the complete guidance for each module of the project was taken under study.

2.1 BUSINESS INTELLIGENCE

Business intelligence (BI) is crucial for organizations due to its ability to transform raw data into actionable insights, enabling informed decision-making and improved business performance. Here are the key reasons why BI is important:

Health Care:

- BI helps analyze patient data for personalized treatment plans and improved healthcare outcomes.
- Hospitals use BI to streamline resource allocation, such as managing staff schedules, optimizing equipment usage, and reducing wait times.
- BI tools detect unusual patterns in billing or insurance claims, reducing fraud and ensuring compliance with regulations.

Retail & E-Commerce:

- BI enables retailers to understand buying patterns, preferences, and trends, leading to personalized marketing campaigns.
- Businesses use BI to forecast demand, optimize stock levels, and prevent overstocking or stockouts.
- Provides insights into competitor pricing, market trends, and customer price sensitivity, allowing dynamic pricing strategies.

Financial Services:

- BI identifies and evaluates financial risks, including credit defaults, market fluctuations, and fraud detection.
- Financial institutions use BI to segment customers based on spending habits, income levels, and investment preferences for targeted services.
- Banks and investment firms track financial metrics and KPIs in real-time, enabling swift adjustments to strategies.

2.2 STREAMLIT

Streamlit is an open-source framework designed to facilitate the creation of web applications for machine learning and data science projects. It's popular because it allows developers to quickly build interactive and data-driven web interfaces with minimal code. Here are some of the main uses and reasons for its popularity:

Data Visualization:

- ☐ Create interactive dashboards and visualizations.
- Display plots from libraries like Matplotlib, Plotly, Altair, and more.

Machine Learning Applications:

- Deploy machine learning models as web applications for easy access.
- Provide interactive interfaces for model input and output visualization.

Prototyping:

- Quickly prototype ideas and see results without extensive frontend development.
- Iterate rapidly by modifying code and seeing changes in real-time.

2.3 PANDAS

Pandas is a popular open-source Python library used for data manipulation and analysis. It provides high-performance, easy-to-use data structures and tools for working with structured data, such as tabular data, time series, and heterogeneous data. Some of the key features and uses of Pandas include:

1. DATA STRUCTURES**2. DATA LOADING AND STORAGE****3. DATA EXPLORATION AND CLEANING****4. DATA MANIPULATION AND TRANSFORMATION****2.4 LINEAR REGRESSION**

Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables. The simplest form, called simple linear regression, involves one dependent variable (Y) and one independent variable (X). Multiple linear regression involves one dependent variable and two or more independent variables.

□ Predictive Modelling:

Predicting outcomes such as sales forecasts, stock prices, and other continuous variables based on predictor variables.

□ Understanding Relationships:

Determining the strength and type of relationship between dependent and independent variables. For example, understanding how changes in marketing impact sales.

□ Trend Analysis:

Identifying and understanding trends in data over time.

□ Risk Management:

Quantifying the risk or effect of variables on a particular outcome, such as credit scoring in finance.

□ Optimizing Processes:

In operations and manufacturing, linear regression can be used to optimize processes by understanding the impact of different factors on production efficiency.

2.5 DECISION TREE REGRESS

K-Means is a popular unsupervised machine learning algorithm used for clustering tasks, where the goal is to group data points into distinct clusters based on their features. Unlike supervised learning models, K-Means does not require labeled data and works by minimizing the variance within clusters.

- **Cluster Centroids:**
K-Means identifies K cluster centroids and assigns each data point to the nearest centroid, iteratively refining the cluster boundaries.
- **Distance-Based Grouping:**
It uses distance metrics, such as Euclidean distance, to measure the similarity between data points and cluster centroids.
- **Iterative Optimization:**
The algorithm alternates between assigning data points to clusters and updating centroid positions until convergence.

2.6 ARIMA

ARIMA (AutoRegressive Integrated Moving Average) is a popular statistical method used for time series forecasting. It combines three key components:

AutoRegressive (AR) part: This component uses the relationship between an observation and a number of lagged observations (previous time steps) to model the time series.

Integrated (I) part: This involves differencing the time series data to make it stationary, which means that the mean and variance of the series are constant over time.

Moving Average (MA) part: This component models the relationship between an observation and a residual error from a moving average model applied to lagged observations.

3. Architecture

3.1 INTRODUCTION

This section provides an outline of the project, its integral parts and major data objects involved and their relationship that flows in the system. It gives us basic understanding about system architecture, how the data objects interact with the system and the overall workflow. The overall running process and the data and interface description are briefly described in this section below.

3.2 SYSTEM ARCHITECTURE

In order to build a scalable, efficient and maintainable software system, layered architecture was built for separation of concern, scalability and maintenance. Each layer is independent of each other as they are configured separately for each process. If need arises, these layers can be replaced by various implementations.

Metro Data

	DateTime	Temperature	LowTemperature	HighTemperature	Humidity
0	2024-07-30 12:00:00	None	33.6	34.4	36.44
1	2024-07-30 13:00:00	34.32	34.1	34.7	34.62
2	2024-07-30 14:00:00	34.38	33.9	34.8	34.43
3	2024-07-30 15:00:00	33.9	33.5	34.2	35.5
4	2024-07-30 16:00:00	33.05	32.8	33.3	35.22

Insects Data

	DateTime	NumOfInsect	NewCapture	Reviewed	Event	Date
0	2024-07-05 11:31:01	0	0	Si	None	2024-07-05
1	2024-07-06 03:02:01	0	0	Si	None	2024-07-06
2	2024-07-07 03:04:01	0	0	Si	None	2024-07-07
3	2024-07-08 03:01:01	0	0	Si	None	2024-07-08
4	2024-07-08 03:10:00	0	0	Si	Cleaning	2024-07-08

4. Functional and Data Description

4.1 INTRODUCTION

This section describes the main modules of this application, sub-modules and their components that are implemented in order to achieve required results.

4.2 DATA LOADING

In this section the data from the dataset is loaded. We first run the streamlit and pmdarima after it the dataset is loaded.

4.2.1 DATASET OVERVIEW:

We have two types of datasets namely Metro and Insects. So we are comparing the common points in both the data and then finding the results.

4.3 MISSING DATA

In this section the missing data is computed and cleaned using the necessary steps possible.

```
# Set the Streamlit page title and configuration
st.set_page_config(page_title='Insects vs Weather Analysis', page_icon='🐞')

# Helper Functions
def load_and_clean_metro(files):
    dfs = []
    for file in files:
        df = pd.read_excel(file, skiprows=0)
        df.columns = df.columns.str.strip()
        expected_columns = {
            "Dati meteo storici": "DateTime",
            "Unnamed: 1": "Temperature",
            "Unnamed: 2": "LowTemperature",
            "Unnamed: 3": "HighTemperature",
            "Unnamed: 4": "Humidity"
        }
        df = df.rename(columns={col: expected_columns.get(col, col) for col in df.columns})
        df['DateTime'] = pd.to_datetime(df['DateTime'], errors='coerce', dayfirst=True)
        df = df.dropna(subset=['DateTime'])
        numeric_columns = ["Temperature", "LowTemperature", "HighTemperature", "Humidity"]
        for col in numeric_columns:
            if col in df.columns:
                df[col] = df[col].str.replace(',', '.').astype(float)
        dfs.append(df)
    return pd.concat(dfs, ignore_index=True)

def load_and_clean_capture(files):
    dfs = []
    for file in files:
        df = pd.read_excel(file, skiprows=0)
        df.columns = df.columns.str.strip()
        df = df.rename(columns={
            "Grafico delle catture": "DateTime",
            "Unnamed: 1": "NumOfInsect",
            "Unnamed: 2": "NumOfInsect",
            "Unnamed: 3": "NumOfInsect",
            "Unnamed: 4": "NumOfInsect"
        })
```

```

df = df.dropna(subset=['DateTime'])
for col in ["NumOfInsect", "NewCapture"]:
    if col in df.columns:
        df[col] = pd.to_numeric(df[col], errors='coerce').fillna(0)
dfs.append(df)
return pd.concat(dfs, ignore_index=True)

# Visualization Functions
def plot_captures(captures_per_day):
    captures_per_day['Date'] = pd.to_datetime(captures_per_day['Date']).dt.strftime('%Y-%m-%d')
    fig = go.Figure()
    fig.add_trace(go.Bar(x=captures_per_day['Date'], y=captures_per_day['TotalInsects'], name='Total Insects', marker_color='blue'))
    fig.add_trace(go.Bar(x=captures_per_day['Date'], y=captures_per_day['NewCaptures'], name='New Captures', marker_color='orange'))
    fig.update_layout(
        title='Total Insects and New Captures per Day',
        xaxis_title='Date',
        yaxis_title='Count',
        barmode='group',
        xaxis=dict(tickangle=-45),
        template='plotly_white',
        legend=dict(title='Legend')
    )
    st.plotly_chart(fig)

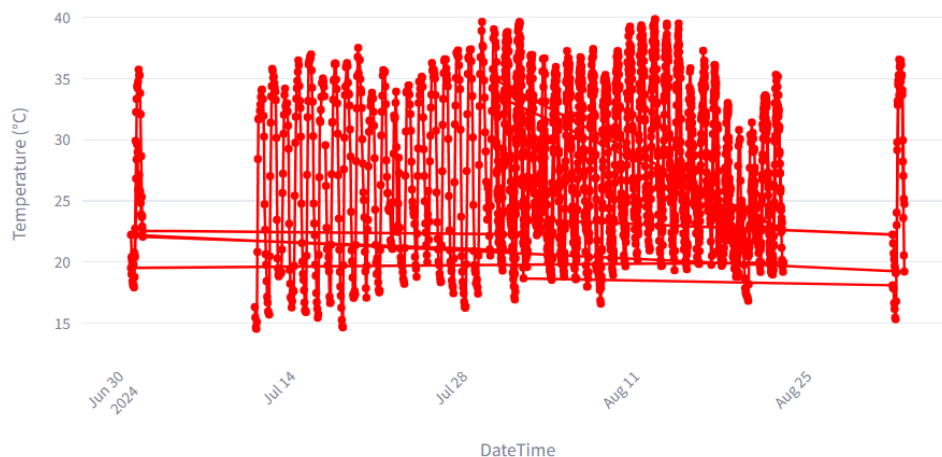
def plot_metro_data_filtered(metro_data):
    # Convert DateTime column to pandas datetime format
    metro_data['DateTime'] = pd.to_datetime(metro_data['DateTime'])

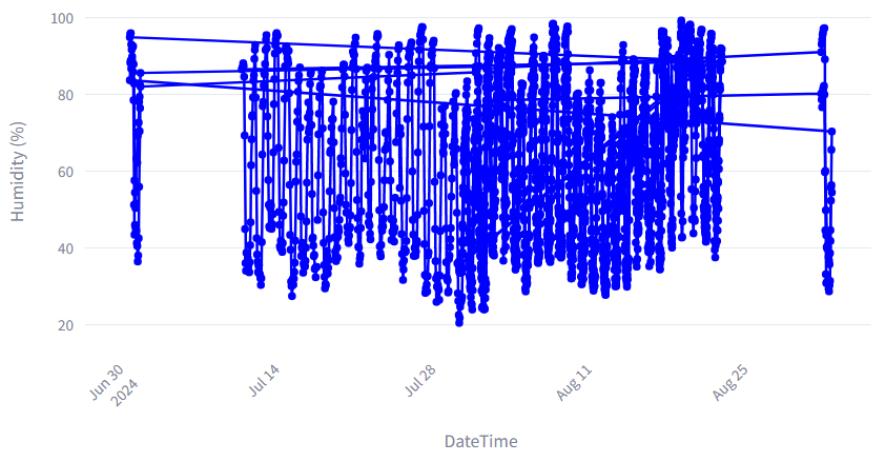
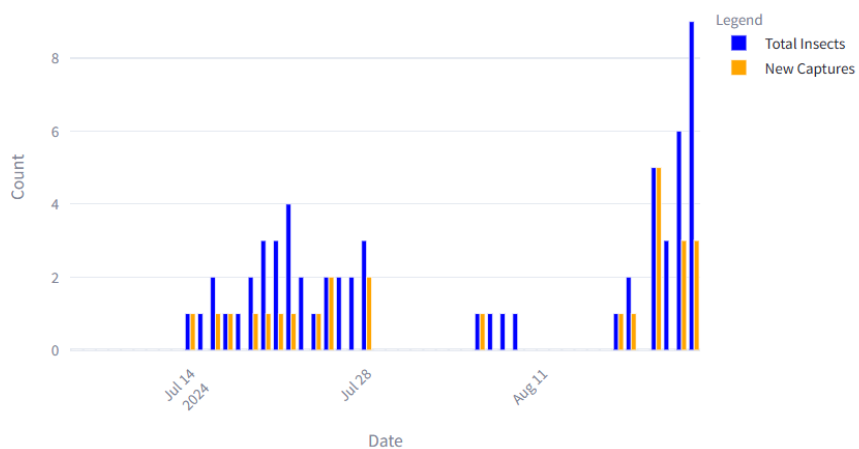
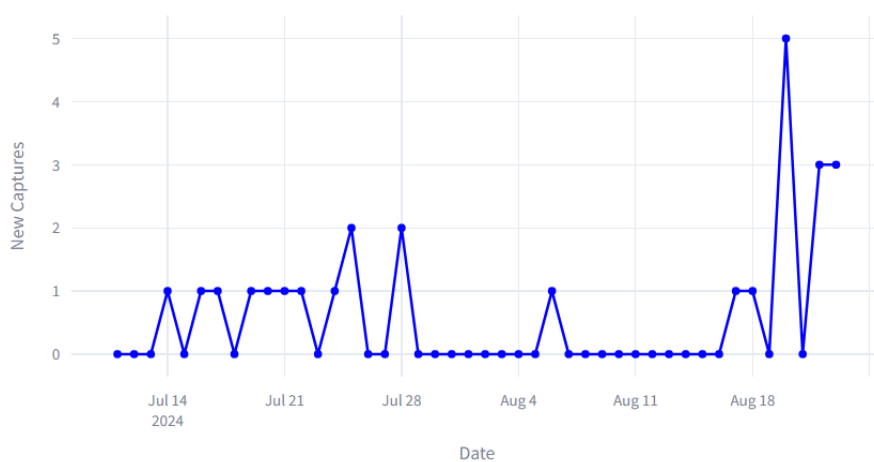
    # Filter data for July to September
    metro_data_filtered = metro_data[metro_data['DateTime'].dt.month.isin([7, 8, 9])]

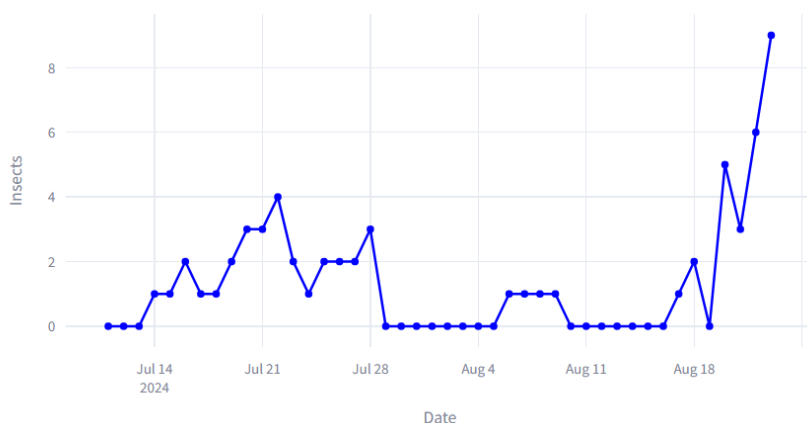
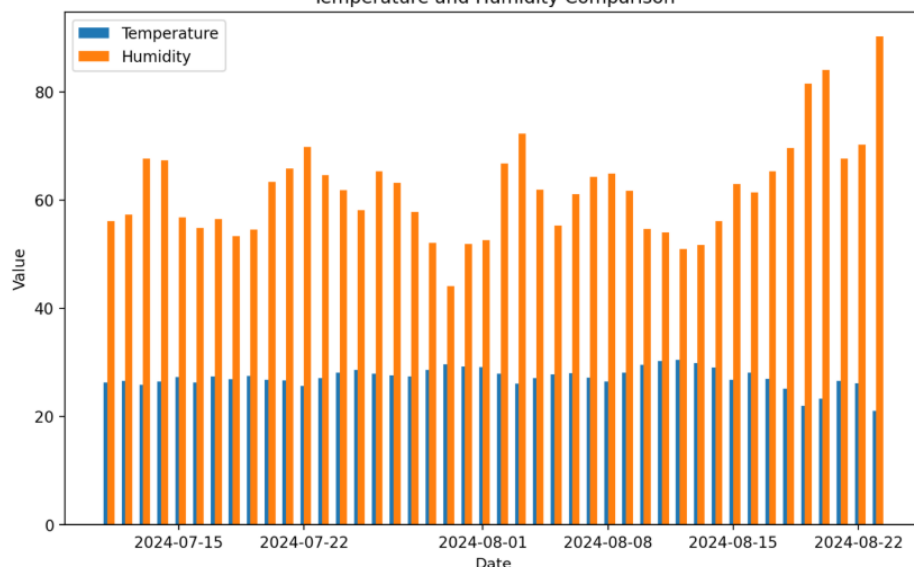
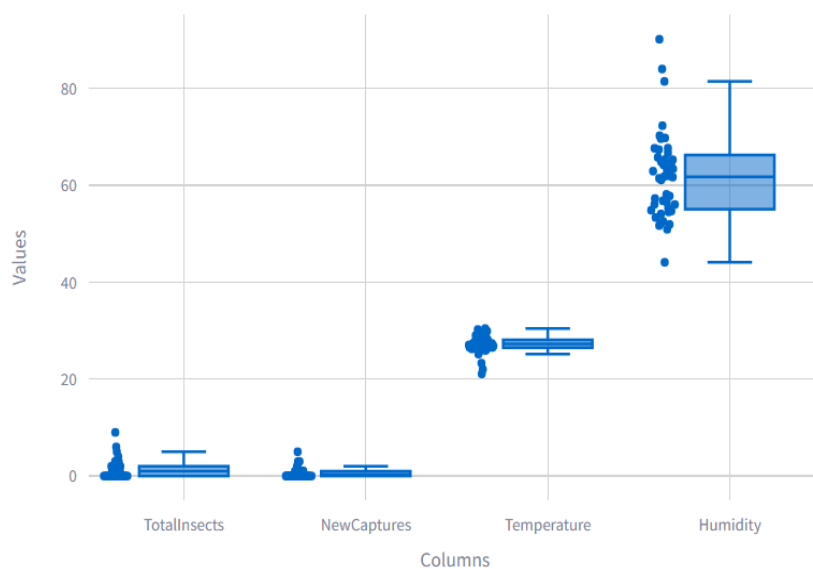
```

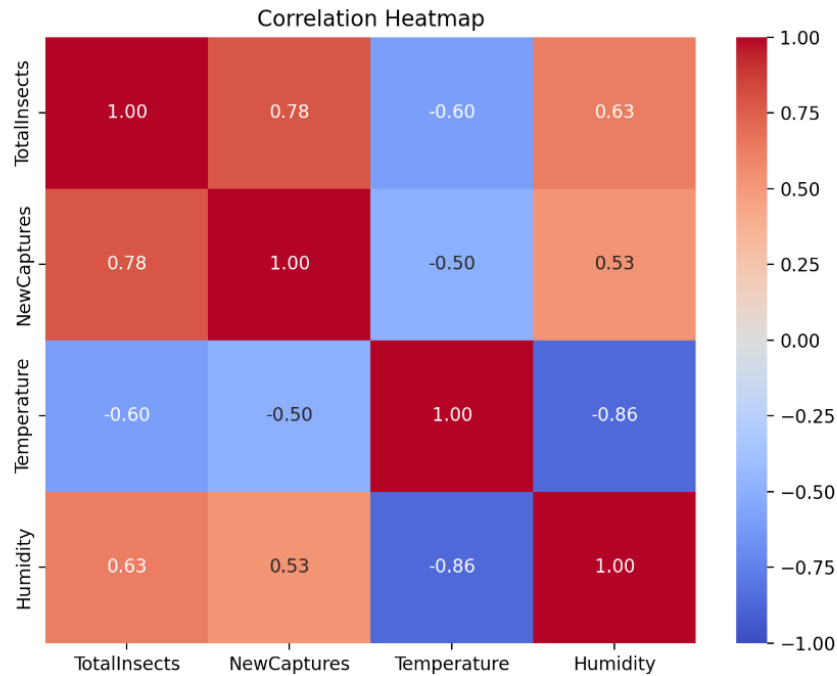
4.4 GRAPHICAL REPRESENTATION:

Temperature Over Time (July to September)



Humidity Over Time (July to September)[Download plot as a png](#)**Total Insects and New Captures per Day****Number of Captured Insects Over Time**[Zoom](#)

Number of Insects Over Time**Temperature and Humidity Comparison****Boxplots of Selected Columns**

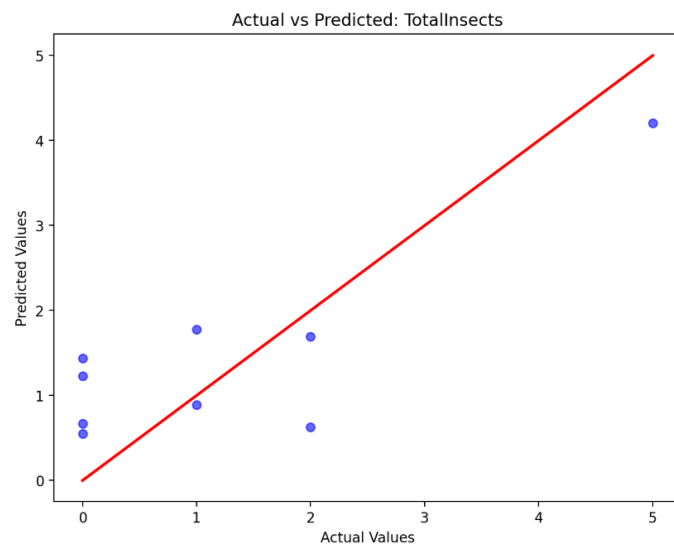


4.5 LINEAR REGRESSION

In this section the data was shaped into the following data: The predicted scores and the actual scores:

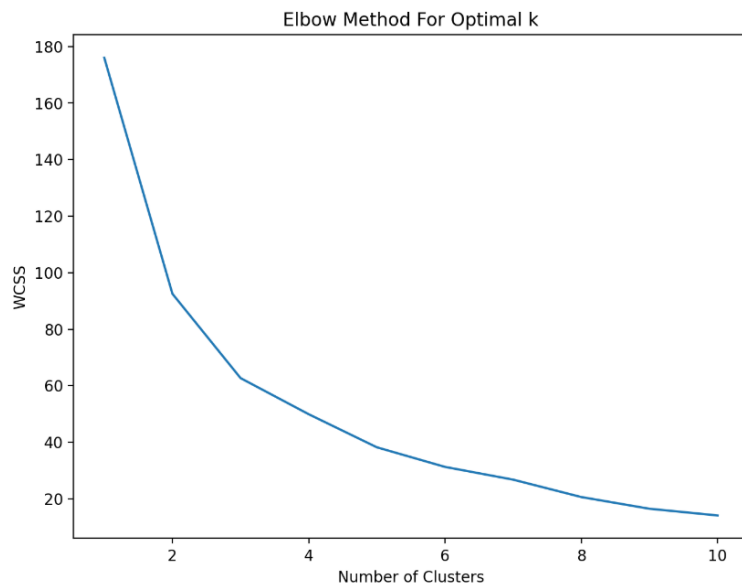
MSE: 0.84

R-Squared: 0.65



4.6 CLUSTERING

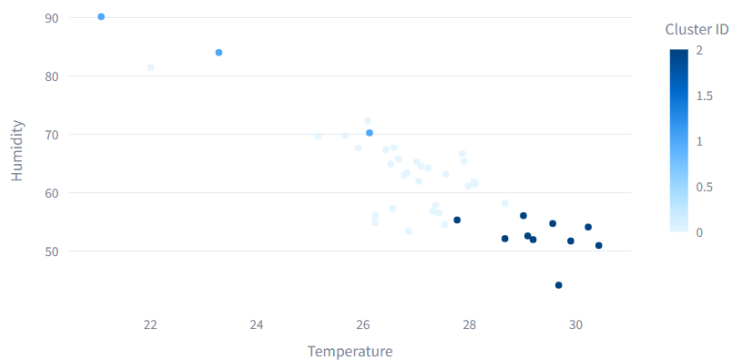
Elbow Method for Optimal Number of Clusters ⇄



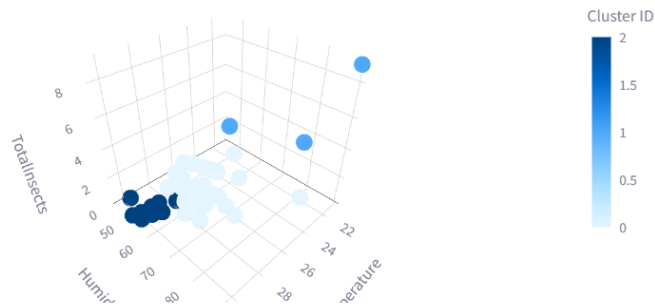
Clustered Data Preview

	Date	TotalInsects	NewCaptures	Temperature	Humidity	Cluster
0	2024-07-11	0	0	26.23	56.0617	0
1	2024-07-12	0	0	26.5579	57.2738	0
2	2024-07-13	0	0	25.9079	67.6421	0
3	2024-07-14	1	1	26.4271	67.3675	0
4	2024-07-15	1	0	27.3113	56.8117	0

Cluster Visualization (Temperature vs. Humidity) - 3 Clusters



3D Cluster Visualization - 3 Clusters



4.7 ARIMA

ADF Statistic: -1.0500075531050543

P-value: 0.7345913324991543

ARIMA Model Summary

SARIMAX Results

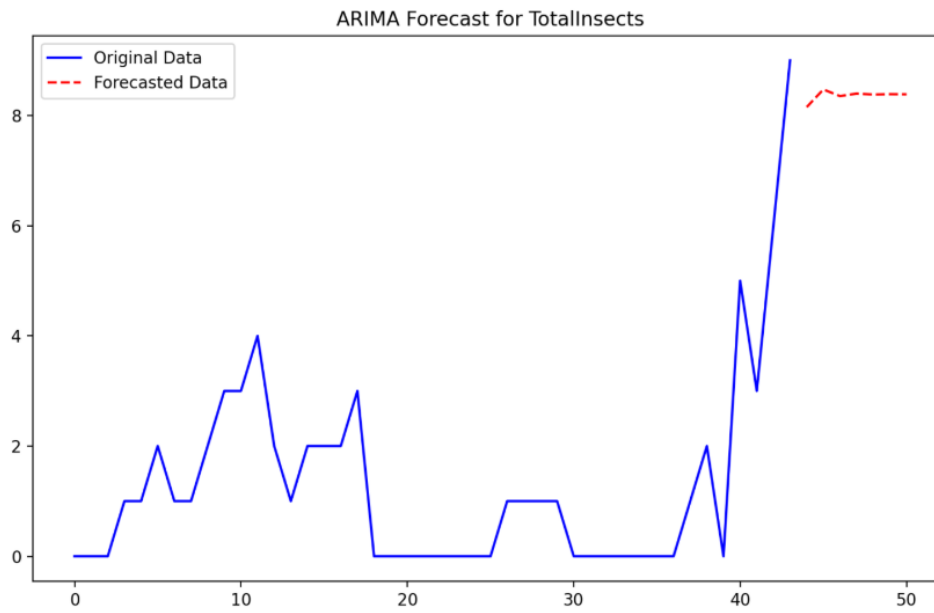
```
=====
Dep. Variable:      TotalInsects   No. Observations:      44
Model:             ARIMA(1, 1, 1)  Log Likelihood         -71.480
Date:              Wed, 15 Jan 2025  AIC                        148.961
Time:              17:54:47         BIC                        154.244
Sample:            0               HQIC                       150.909
                  - 44
Covariance Type:    opg
=====
```

	coef	std err	z	P> z	[0.025	0.975]
ar.L1	-0.3779	0.421	-0.898	0.369	-1.203	0.447
ma.L1	0.0718	0.541	0.133	0.894	-0.989	1.132
sigma2	1.6231	0.219	7.424	0.000	1.195	2.052

```
=====
Ljung-Box (L1) (Q):      0.03   Jarque-Bera (JB):      2
Prob(Q):                 0.86   Prob(JB):
Heteroskedasticity (H):   3.75   Skew:
Prob(H) (two-sided):      0.02   Kurtosis:
=====
```

Forecasted Values for Next 7 Days

	predicted_mean
44	8.1513
45	8.472
46	8.3508
47	8.3966
48	8.3793
49	8.3859
50	8.3834



4.8 LIBRARIES

```
import streamlit as st
import pandas as pd
import numpy as np
import plotly.express as px
import plotly.graph_objects as go
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from statsmodels.tsa.arima.model import ARIMA

from statsmodels.tsa.stattools import adfuller # Make sure to import adfuller
```

5. Conclusion

. 5.1 SUMMARY

In this report, we addressed the missing values in two primary datasets: the **Metro Data**, which included information related to temperature, and the **Captured Insects Data**, which detailed insect capture data with timestamps. We performed exploratory data analysis (EDA) and applied predictive modelling techniques—Linear Regression, ARIMA, and K-Means clustering—to derive meaningful insights and make predictions.

Handling Missing Values

The missing values in the datasets were handled using appropriate imputation techniques. We used pandas' built-in functions to replace missing values based on domain knowledge and the distribution of the data. This ensured that the datasets were complete and ready for subsequent analysis.

Exploratory Data Analysis (EDA)

EDA was carried out to understand the distribution, relationships, and patterns within the datasets. Libraries like pandas, matplotlib, seaborn, and plotly were used for visualization. Key insights from the EDA included:

- Temperature trends over different time periods.
- Seasonal variations in temperature and insect capture.
- Relationships between temperature and insect capture numbers.
- Variability in the captured insects based on time and environmental factors.
-

Predictive Modelling

Linear Regression

Linear regression was applied to predict the number of insects captured based on temperature and other environmental factors. The performance metrics obtained were:

- **MSE:** 0.84
- **R-Squared:** 0.65

These results indicated that the linear regression model was able to capture some of the variability in insect capture but also highlighted the need for more complex models to fully account for non-linear patterns.

K-Means Clustering

K-Means clustering was used to segment the data into different groups based on temperature and insect capture characteristics. The results showed:

- **Number of Clusters:** 3
- The clustering model helped identify distinct patterns in temperature and insect captures, revealing possible correlations between environmental conditions and insect populations.

ARIMA

ARIMA was employed for time series forecasting to predict future insect captures based on historical data. The stationarity test results were:

- **ADF Statistic:** -1.0500075531050543
- **P-value:** 0.7345913324991543

These metrics suggested that the time series data might not have been strongly stationary, but ARIMA still provided reasonably accurate short-term predictions of insect captures.

Summary

The analysis and modelling conducted in this report demonstrated the utility of predictive techniques in understanding the relationship between temperature and insect capture over time. While **Linear Regression** provided a baseline performance, **K-Means clustering** helped segment the data into meaningful groups, revealing more nuanced patterns. **ARIMA** was effective for time series forecasting, offering insights into potential future trends.

The successful handling of missing values, coupled with comprehensive EDA, provided a solid foundation for building reliable models. Future work could explore more advanced techniques, such as Random Forest or Neural Networks, to enhance predictive accuracy. Additionally, refining the features and exploring different configurations of ARIMA may yield better results.

In conclusion, the findings of this report can be utilized to inform decision-making in areas like environmental management and pest control, allowing for better predictions and optimized resource allocation based on temperature patterns and insect behaviors.

This adjusted conclusion takes into account the context of your dataset, specifically related to temperature and insect capture, while reflecting the application of **Linear Regression**, **ARIMA**, and **K-Means Clustering**.

6. References

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